Band-1 Computational Experiment: Bounded Prime Gaps

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Bounded Prime Gaps (Maynard–Tao / Polymath)

Honesty stamp. This note (i) ledgers the literature on the unconditional bound

$$H_1 := \liminf_{n\to\infty}(p_{n+1} - p_n),$$

and (ii) runs a **finite-range computational illustration** counting consecutive prime pairs with gap $\leq H$. It does **not** prove any liminf theorem; it does **not** claim twin primes ($H_1 = 2$) or Polignac solved; it does **not** claim a new record bound on H_1 without a real proof. Finite observation $\neq$ infinitude.

Band-4 callout. Gap = 2 (the **twin-prime conjecture**) remains **Band-4 / OPEN**. Even under the generalized Elliott–Halberstam conjecture, the Maynard–Tao / Polymath sieve reaches only $H_1 \leq 6$ (parity obstruction), not 2.

Abstract

Literature (published standing record): Polymath8b proved

$$H_1 \leq 246$$

unconditionally (*Research in the Mathematical Sciences* **1** (2014), Article 12; arXiv:1407.4897; DOI 10.1186/s40687-014-0012-7), via Maynard–Tao multidimensional sieve weights and the Bombieri–Vinogradov theorem, using the shortest admissible 50-tuple of diameter 246.

2026 preprints (ledger-noted, not elevated here): Stadlmann, arXiv:2608.31126, claims $H_1 \leq 240$; a separate ePrint claim of $H_1 \leq 236$ is recorded as **unverified / not elevated**.

Box illustration: among the $\pi(10^7) = 664{,}579$ primes in $[2, 10^7]$, all 664,578 consecutive gaps satisfy gap ≤ 246 (indeed max gap = 154); 58,980 consecutive twin pairs (gap = 2); counts for $H \in \{2, 6, 12, 70, 246\}$ below. Wall ≈ 0.12 s. **Illustration only.**

Problem statement

Bounded-gap theorem (Zhang / Maynard / Polymath). There exists an absolute constant C such that $p_{n+1} - p_n \leq C$ for infinitely many n. The best widely-cited **published** unconditional C is 246.

This experiment. Ledger the machine (Zhang → Polymath8a → Maynard → Polymath8b) and any later claimed improvements with status tags; then sieve $[2, N]$ and count consecutive pairs with $p_{n+1} - p_n \leq H$ for chosen H (default $H = 246$).

Literature ledger (not re-proved on box)

Unconditional history for H_1

Year	Bound	Authors	Status
2013/14	7×10^7	Zhang, *Ann. of Math.* 179 (2014)	published (first finite)
2013/14	4680	Polymath8a, *Algebra Number Theory* 8 (2014)	published
2013/15	600	Maynard, *Ann. of Math.* 181 (2015)	published
2014	246	**Polymath8b**, *Res. Math. Sci.* 1 (2014) Art. 12	**published standing**
2026	240	Stadlmann, arXiv:2608.31126	**preprint — not elevated**
2026	236	Song–Yue (claimed), eprint.iacr.org/2026/1893	**unverified — not elevated**

Primary standing citation: D.H.J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Research in the Mathematical Sciences **1** (2014), Article 12 (arXiv:1407.4897; DOI 10.1186/s40687-014-0012-7).

Upstream: - Y. Zhang, *Bounded gaps between primes*, Ann. of Math. **179** (2014), 1121–1174. - J. Maynard, *Small gaps between primes*, Ann. of Math. **181** (2015), 383–413.

Conditional (context only)

Assumption	Bound	Source
Elliott–Halberstam (EH)	$H_1 \leq 12$	Maynard / Polymath8b
Generalized EH (GEH)	$H_1 \leq 6$	Polymath8b (sieve limit)

Still not twin primes: GEH $\Rightarrow H_1 \leq 6$ does **not** prove $H_1 = 2$.

Method (box)

1. **Sieve of Eratosthenes** on $[0, N]$ with $N = 10^7$; extract prime list.
2. **Consecutive gaps** $g_n = p_{n+1} - p_n$; count $\{n : g_n \leq H\}$ for $H \in \{2, 6, 12, 70, 246\}$.
3. **Histogram** of small gaps; sample pairs; unit tests ($\pi(10^6) = 78498$; twins with first < 100; known gaps after 101, 113).
4. **No claim** that finite counts imply liminf bounds.

Results (numbers!)

Unit tests

`PASS`: primes ≤ 30; $\pi(10^6) = 78498$; gap after 101 is 2; gap after 113 is 14; 8 twin pairs with first prime < 100.

Illustration ($N = 10^7$, $H_{\text{main}} = 246$)

Quantity	Value
N	10,000,000
$\pi(N)$	664,579
Consecutive gaps	664,578
Gaps ≤ 246	664,578 **(100%)**
Gaps > 246	0
Min gap	1 (only $(2, 3)$)
Max gap in range	154
Mean gap	15.047
Consecutive twins (gap = 2)	58,980
Gaps = 4	(see `results.json` histogram)
Gaps = 6	(see `results.json`)
Wall	≈ 0.12 **s** (sieve ≈ 0.055 s + scan ≈ 0.059 s)

Counts of consecutive gaps $\leq H$

H	$\#\{n : p_{n+1} - p_n \leq H\}$ in $[2, 10^7]$
2	58,981 (includes the unique gap 1)
6	217,589
12	379,885
70	662,158
246	664,578

Interpretation. At $N = 10^7$, the largest consecutive gap is $154 < 246$, so every gap is ≤ 246. That is a finite-range fact (and matches known gap tables at this scale). It **illustrates** that gaps ≤ 246 are abundant; it **does not** prove that such gaps occur infinitely often — that is Polymath8b's theorem, not this sieve.

Sample consecutive pairs with gap ≤ 246 (first few)

p	q	gap
2	3	1
3	5	2
5	7	2
7	11	4

p	q	gap
11	13	2
...		

Why Band-1 vs Band-4

- **Band-1 (this note):** literature ledger hygiene for the Maynard–Tao / Polymath machine; finite-range illustration that gaps $\leq H$ are common; methods certificate (sieve + gap census).
- **Band-4:** prove $H_1 = 2$ (twin primes) or any specific even Polignac number. Still **OPEN**; **not claimed**. GEH only reaches 6.

Reproducibility

- Code: `/workspace/experiments/band1/bounded_gaps/`
- Run: `python3 bounded_gaps.py`
- Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_bounded_gaps.log`

Conclusion

Literature: published unconditional standing bound remains $H_1 \leq 246$ (Polymath8b 2014). 2026 preprints claiming 240 / 236 are **noted, not elevated**. **Box:** in $[2, 10^7]$, 58,980 consecutive twin pairs and 100% of consecutive gaps ≤ 246 (max = 154) — illustration only. **Twin primes / Polignac: Band-4 OPEN. No new liminf record claimed.**